

15.095 MACHINE LEARNING UNDER A MODERN OPTIMIZATION LENS

Multimodal Alpha Forecasting and Robust Portfolio Optimization

Presented by

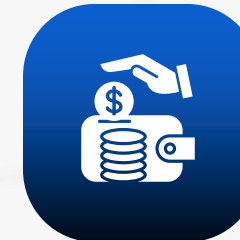
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The Challenge - Signal in the Noise



LOW SIGNAL-TO-NOISE RATIO

Daily returns are chaotic and dominated by noise.



COMPLEX INTERACTIONS

Returns reflect non-linear relationships between macro conditions and firm specifics.



THE GOAL

Integrate **multimodal** data (Price, Fundamentals, Text) to generate robust forecasts for **prescriptive** portfolio optimization.

Data



Integrating structured tabular data with unstructured text data

Structured Data



Market Price

Daily OHLCV
prices from
Yahoo Finance



Fundamentals

Profitability,
growth, and
valuation ratios
from **WRDS**



Macro

GDP, VIX, and
Treasury yields
from **FRED**



Unstructured Data

Date string	Article_title string	Stock_symbol string	Url string
2020-06-05...	Stocks That Hit 52-Week...	A	https://www.benzinga.com/s-that-hit-52-week-high/
2020-06-03...	Stocks That Hit 52-Week...	A	https://www.benzinga.com/s-that-hit-52-week-high/
2020-05-26...	71 Biggest Movers From...	A	https://www.benzinga.com/biggest-movers-from-fr/
2020-05-22...	46 Stocks Moving In...	A	https://www.benzinga.com/stocks-moving-in-friday/

News/Text

Headlines and
summaries from
FNSPID

Feature Engineering



We leverage the **FinBERT** model to process unstructured news headlines and summary

Date string	Article_title string	Stock_symbol string	Url string	Publisher string
2020-06-05 06:30:54 U...	Stocks That Hit 52-Week Highs On Friday	A	https://www.benzinga.com/news...	Benzinga Insights
2020-06-03 06:45:20 U...	Stocks That Hit 52-Week Highs On Wednesday	A	https://www.benzinga.com/news...	Benzinga Insights
2020-05-26 00:30:07 U...	71 Biggest Movers From Friday	A	https://www.benzinga.com/news...	Lisa Levin
2020-05-22 08:45:06 U...	46 Stocks Moving In Friday's Mid-Day Session	A	https://www.benzinga.com/news...	Lisa Levin
2020-05-22 07:38:59 U...	B of A Securities Maintains Neutral on Agilent	A	https://www.benzinga.com/news...	Vick Meyer
2020-05-22 07:23:25 U...	CFRA Maintains Hold on Agilent Technologies, Inc.	A	https://www.benzinga.com/news...	vishwanath@benzinga.com
2020-05-22 05:36:20 U...	UBS Maintains Neutral on Agilent Technologies, Inc.	A	https://www.benzinga.com/news...	vishwanath@benzinga.com
2020-05-22 05:07:04 U...	Agilent Technologies shares are trading high	A	https://www.benzinga.com/wiim...	Benzinga Newsdesk
2020-05-22 04:37:59 U...	Wells Fargo Maintains Overweight on Agilent	A	https://www.benzinga.com/news...	vishwanath@benzinga.com
2020-05-22 04:06:17 U...	10 Biggest Price Target Changes For Friday	A	https://www.benzinga.com/anal...	Lisa Levin

Sentiment Scores

Classifying news as Positive, Negative, or Neutral.

Aggregated to daily mean/extremal scores.

Text Embeddings

Extracting 768-dimensional contextual vectors from news.

Dimensionality Reduction

Using PCA to compress embeddings to 64 components for efficient modeling.

Baseline Model



Single Rule Baseline

- Built simple forecasting rules applied uniformly across all stocks:
 - *Persistence*: tomorrow's return = today's return
 - *Moving Averages (MA)*: predict using past 3 – 63-day averages
 - *Exponential MAs*: smoother momentum via decaying weights
 - *Seasonal Rules (S1–S7)*: same-day-last-year, same-month-last-year, month-of-year mean, day-of-week mean, long-horizon rolling means



Hybrid Baseline

- *Per-Ticker*: best rule per stock
- *Per-Month*: best rule per calendar month
- *Per-Day-of-Week*: best rule per weekday



Key Insights

- Single rule baseline remain very close to the 50% benchmark
- Allowing different segments to select their own best rule increases in-sample accuracy but with limited true generalization
- Because the per-ticker hybrid achieved the strongest in-sample accuracy, its selection patterns guided which features to emphasize in the advanced models

Selection Frequency by Rule (Per-Ticker Hybrid)



From Baseline to ML Models

Feature Sets Evaluated

- Baseline-derived signals only
- Core stock structure only (Day-of-week, month, sector, subsector, volume)
- Combined feature set

Models Tested Across All Feature Sets

- Linear Regression
- Neural Network with ticker embeddings
- XGBoost
- CatBoost
- LightGBM

Feature Importance & Ablation

- Gain-based feature importance
- Ran systematic ablations
 - Different combinations of features
 - Top-K ranked features

Key Finding

Across all model classes and feature subsets, the most consistently predictive signals were:
Day-of-Week + Month-of-Year + Sector (DMS)

Machine Learning Models

Starting point:

Best structural baseline: DMS = Day-of-Week + Month + Sector



Feature Groups Built on Top of DMS

Fundamentals
Macro variables
Sentiment & Text



Best LightGBM Models

Best Validation DA: LightGBM – DMS + selected fundamentals

Best Test DA: LightGBM – DMS + mean_sentiment

Robust CVaR Portfolio Optimization



We transform **raw predictions** into a portfolio by **maximizing returns** while strictly controlling for **Conditional Value-at-Risk**

Optimization Objective

Conditional Value-at-Risk (CVaR) tells us how bad the losses can get on average in the **worst 5%** of scenarios

$$CVaR_{\alpha}(w) = v + \frac{1}{((1 - \alpha)T)} * \sum_{t=1}^T z_t$$
$$z_t \geq 0, z_t \geq -(r_t^T w) - v$$

The optimizer balances **predicted return**, **downside risk (CVaR)**, and **transaction costs** to choose the most attractive portfolio **weights**

$$\max_w$$
$$\hat{\mu}_t^T w - \lambda \cdot CVaR_{0.95}(w) - TC \cdot \|w - w_{t-1}\|_1$$

Trading Constraints

Long-only and fully invested

$$w_i \geq 0, \sum_i w_i = 1$$

Position Limit

$$w_i \leq 0.02$$

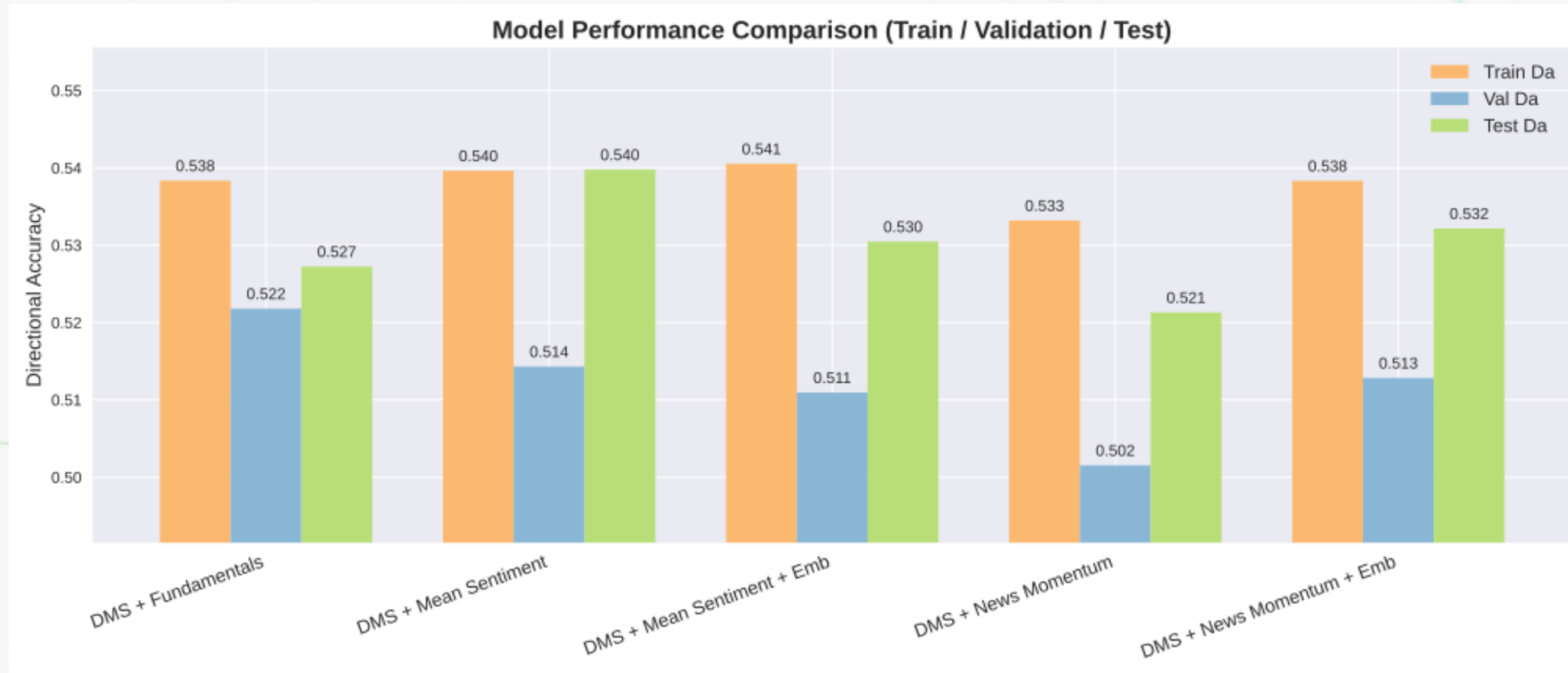
Turnover Limit

$$\|w - w_{t-1}\|_1 \leq 0.10$$

Transaction Cost Penalty

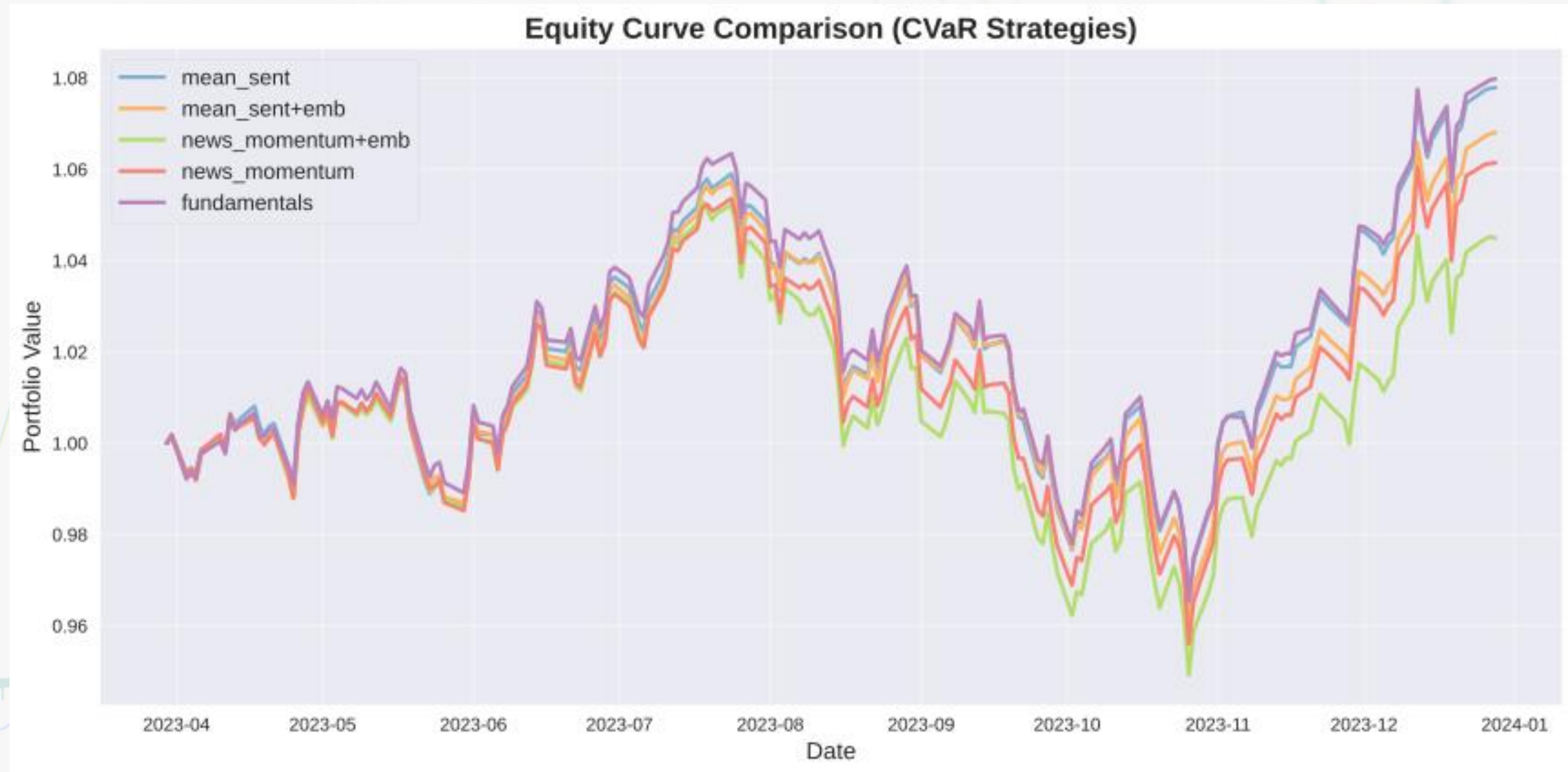
$$TC = 0.0005 \cdot \|w - w_{t-1}\|_1$$

Results - Predicted Return Directional Accuracy



- All models beat random guessing
- **DMS + Mean Sentiment perform best**, showing that **slow-moving seasonality patterns** and **broad news sentiment** provide the most stable signals
- **Text embeddings add noise**, offering little to no improvement and occasionally reducing accuracy due to overfitting

Results - Portfolio Performance under Optimization



Results - Portfolio Performance under Optimization

Feature Set	Sharpe Ratio	Ann. Return	Mean IC
DMS + Fundamentals	1.166	10.84%	0.028
DMS + News Mean Sentiment	1.142	10.58%	0.027
DMS + Sentiment + Embeddings	0.989	9.23%	0.021
DMS + News Momentum + Embeddings	0.654	6.07%	0.015

- **Slow-moving fundamentals** and **aggregate news sentiment** remain the most reliable predictors at a daily horizon.
- Adding text embeddings consistently reduces both Sharpe and IC, indicating that high-dimensional embeddings introduce noise rather than improving predictability.

Conclusions



Noise Dominance

Daily returns are extremely noisy, and even advanced multimodal models only reach ~51–54% directional accuracy.



Less is More

High-dimensional embeddings introduced noise. Simple features (fundamentals, aggregate news sentiment) provided the most reliable signals.



Robustness Matters

Even with weak predictions (DA ~54%), CVaR optimization successfully constructed profitable, risk-controlled portfolios.



Thank You